

AI Sales SQL Analyst

Testing & Validation Report

Project	AI-powered Sales SQL Analyst
Interface	Streamlit
Database	PostgreSQL
LLM Runtime	Ollama + Qwen2.5-Coder 7B
Testing Scope	5 end-to-end business questions

This report documents five representative Streamlit tests covering revenue analysis, customer ranking, time-series analysis, business metrics, and period-over-period comparison.

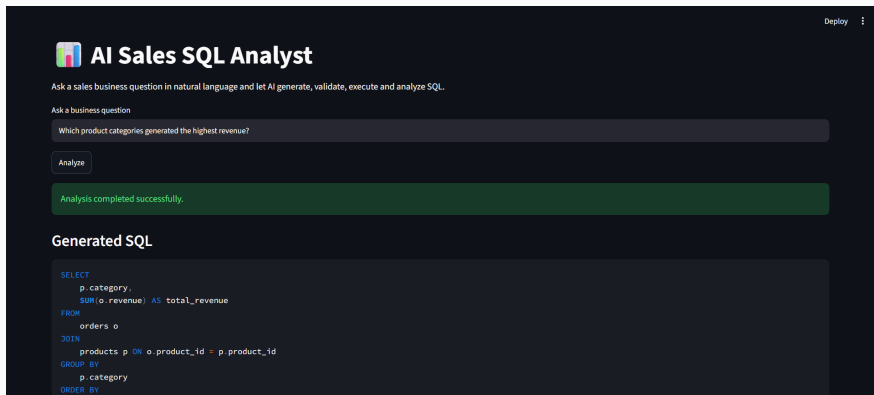
Test 1 — Revenue by Product Category

Business
Question:

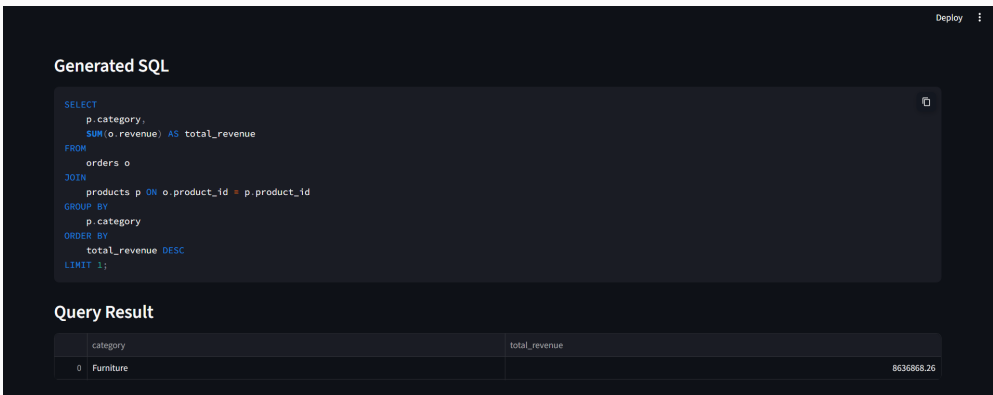
Which product categories generated the highest revenue?

Result: Furniture generated the highest revenue with 8,636,868.26.

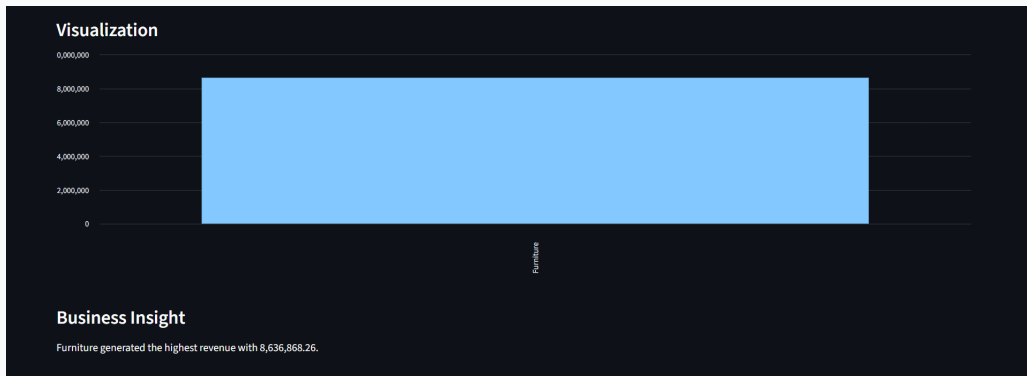
Streamlit Evidence



Question and generated SQL



Generated SQL and query result



Visualization and business insight

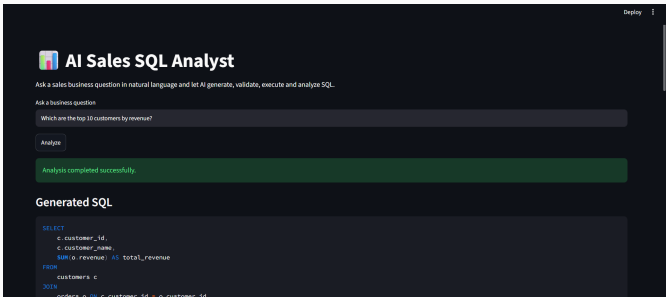
Test 2 — Top 10 Customers by Revenue

Business Question:

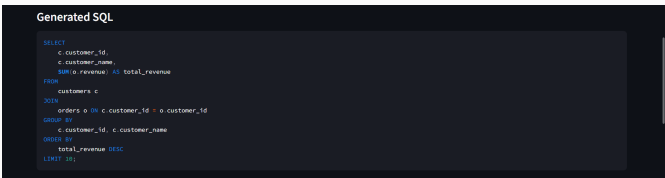
Which are the top 10 customers by revenue?

Result: Customer 638 ranked first among the top 10 customers with revenue of 63,188.72.

Streamlit Evidence



Question and generated SQL

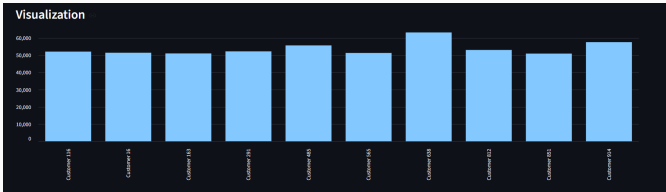


Generated SQL

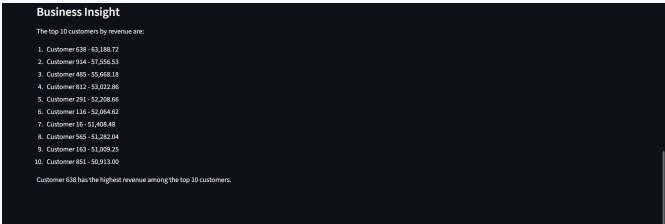
The screenshot shows the 'Query Result' table, which lists the top 10 customers by revenue. The table has three columns: customer_id, customer_name, and total_revenue.

customer_id	customer_name	total_revenue
638	Customer 638	63188.72
914	Customer 914	57556.53
495	Customer 495	56668.19
812	Customer 812	53022.86
291	Customer 291	52208.66
116	Customer 116	52064.62
16	Customer 16	51456.48
565	Customer 565	51282.94
163	Customer 163	51099.25
851	Customer 851	50913

Query result



Visualization



Business insight

Test 3 — Monthly Revenue Trend

Business
Question:

What was the monthly revenue trend in 2024?

Result: May had the highest monthly revenue at 1,379,739.44, while November had the lowest at 1,150,142.99.

Streamlit Evidence

AI Sales SQL Analyst

Ask a sales business question in natural language and let AI generate, validate, execute and analyze SQL.

Ask a business question

What was the monthly revenue trend in 2024?

Analyze

Analysis completed successfully.

Question and generated SQL

Generated SQL

```
SELECT
  EXTRACT(MONTH FROM orders.order_date) AS month,
  SUM(orders.revenue) AS revenue
FROM
  orders
WHERE
  EXTRACT(YEAR FROM orders.order_date) = 2024
GROUP BY
  EXTRACT(MONTH FROM orders.order_date)
ORDER BY
  month;
```

Generated SQL

Query Result

month	revenue
0	1301504.07
1	1293164.07
2	1334110.04
3	1175761.21
4	1379739.44
5	1244841.65
6	1254930.68
7	1311546.29
8	1338245.97
9	1278446.78

Query result

Visualization

Business Insight

The monthly revenue trend in 2024 showed a fluctuating pattern, with the highest revenue in May (1,379,739.44) and the lowest in November (1,150,142.99).

visualization and business insight

Test 4 — Average Order Value by Segment

Business
Question:

What is the average order value for each customer segment?

Result: Small Business had the highest average order value at 1,536.07, while Consumer had the lowest at 1,505.43.

Streamlit Evidence

AI Sales SQL Analyst

Ask a sales business question in natural language and let AI generate, validate, execute and analyze SQL.

Ask a business question

What is the average order value for each customer segment?

Analyze

Analysis completed successfully.

Question and generated SQL

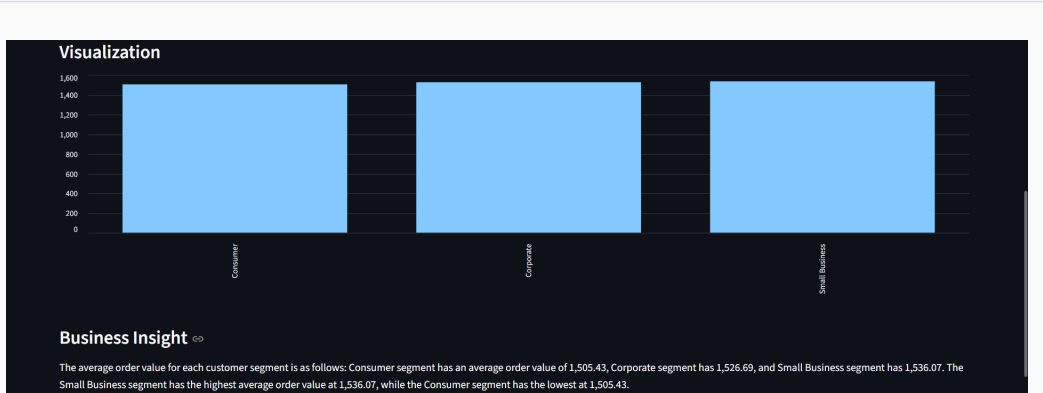
Generated SQL

```
SELECT
  customers.segment,
  SUM(orders.revenue) / COUNT(DISTINCT orders.order_id) AS average_order_value
FROM
  orders
JOIN
  customers ON orders.customer_id = customers.customer_id
GROUP BY
  customers.segment;
```

Query Result

	segment	average_order_value
0	Consumer	1505.4324
1	Corporate	1526.6875
2	Small Business	1536.0669

Generated SQL and query result



Visualization and business insight

Test 5 — Q1 vs Q2 Revenue Decline

Business Question:

Which customer segments had the largest revenue decline in Q2 compared with Q1 2024?

Result: Small Business had the largest revenue decline, with a decrease of 114,404.30.

Streamlit Evidence

AI Sales SQL Analyst

Ask a sales business question in natural language and let AI generate, validate, execute and analyze SQL.

Ask a business question

Which customer segments had the largest revenue decline in Q2 compared with Q1 2024?

Analyze

Analysis completed successfully.

Generated SQL

```
WITH revenue_q1 AS (
  SELECT
    segment,
    SUM(revenue) AS revenue
  FROM
    orders
  JOIN
    customers ON orders.customer_id = customers.customer_id
  WHERE
    order_date BETWEEN '2024-01-01' AND '2024-03-31'
  GROUP BY
    segment
),
revenue_q2 AS (
  SELECT
    segment,
    SUM(revenue) AS revenue
  FROM
    orders
  JOIN
    customers ON orders.customer_id = customers.customer_id
  WHERE
    order_date BETWEEN '2024-04-01' AND '2024-06-30'
  GROUP BY
    segment
)
SELECT
  q1.segment,
  q1.revenue AS revenue_q1,
  q2.revenue AS revenue_q2,
  (q1.revenue - q2.revenue) AS revenue_decline
FROM
  revenue_q1 q1
JOIN
  revenue_q2 q2 ON q1.segment = q2.segment
ORDER BY
  revenue_decline DESC;
```

Generated SQL

Query Result

	segment	revenue_q1	revenue_q2	revenue_decline
0	Small Business	1481967.13	1367562.83	114404.3
1	Consumer	1198512.73	1168461.03	30051.7
2	Corporate	1248298.32	1264318.44	-16020.12

Query Result

Visualization

segment	revenue_q1	revenue_q2
Consumer	1198512.73	1168461.03
Corporate	1248298.32	1264318.44
Small Business	1481967.13	1367562.83

Visualization

Business Insight

The customer segment with the largest revenue decline in Q2 compared to Q1 2024 was Small Business, with a decline of 114,404.30.

Business Insight

Page 1

Testing Summary

The five Streamlit tests below were completed as end-to-end demonstrations of the AI Sales SQL Analyst workflow.

Test	Business Scenario	Status
1	Revenue by product category	PASS
2	Top 10 customers by revenue	PASS
3	Monthly revenue trend in 2024	PASS
4	Average order value by customer segment	PASS
5	Q1 vs Q2 revenue decline by segment	PASS

Capabilities demonstrated

- Natural-language business questions converted into PostgreSQL SQL.
- Schema-aware joins and sales metric calculations.
- SQL execution against PostgreSQL and tabular result display.
- Automatic visualization of query results.
- Business insight generation from query output.
- Advanced period comparison using Q1 and Q2 revenue.

Conclusion: The five representative tests demonstrate the core end-to-end workflow of the AI Sales SQL Analyst in the Streamlit interface.